

Ironhack Final Project Structure

Food Safety Inspection Analytics & Failure-Risk Prediction

German BVL Food Monitoring Data (2023) | Case Study 1: Data Science End-to-End Project

1. Project Goal

Build an end-to-end data science project around a real food-safety business problem: analyze German food-monitoring data, identify patterns associated with non-compliance or failure risk, and develop a predictive classification model that can support risk-based inspection or monitoring decisions.

2. Rubric-Driven Success Targets

Area	Target for an Excellent submission
Business problem	Multiple focused research questions/hypotheses; actionable recommendations.
Data preparation	Identify and justify quality issues; handle nulls, duplicates, unnecessary columns, strings, types and formatting.
EDA & statistics	Univariate, bivariate and multivariate analysis; informative Python plots; inferential statistics where appropriate.
Machine learning	At least 3 appropriate models; justified preprocessing; cross-validation; hyperparameter tuning; appropriate evaluation.
Code quality	Clean project structure; reusable functions in .py files; clear naming; minimal hard-coding; no unused code.
Git/GitHub	At least one meaningful atomic commit per project day; accurate commit messages.
Documentation	Comprehensive README; docstrings for functions; concise comments explaining rationale.
Presentation	Clear story, findings, demo, conclusions, limitations, recommendations and next opportunities.

3. End-to-End Project Pipeline

- **Phase 0 - Project setup:** Conda environment, dependencies, environment.yml, requirements.txt, Git repository, .gitignore and folder structure.
- **Phase 1 - Business understanding:** Define the food-safety problem, stakeholders, objective, research questions/hypotheses and success criteria.
- **Phase 2 - Data understanding:** Inspect the 2023 BVL dataset safely, document fields, row granularity, data types, categories and candidate outcome variables.
- **Phase 3 - Data cleaning & wrangling:** Missing values, duplicates, data types, strings, invalid values, irrelevant fields, outliers and reproducible cleaning functions.
- **Phase 4 - EDA & statistical analysis:** Descriptive statistics, distributions, category comparisons, relationships, food/product patterns, non-compliance patterns and hypothesis tests where justified.
- **Phase 5 - Feature engineering:** Create defensible predictors from the source data while preventing target leakage.
- **Phase 6 - ML preprocessing:** Train/test split, encoding, scaling where required, class-balance assessment and pipelines.
- **Phase 7 - Modeling:** Baseline plus at least three suitable classifiers; cross-validation and hyperparameter tuning.
- **Phase 8 - Evaluation:** Confusion matrix, precision, recall, F1 and other appropriate metrics; compare models and select the simplest effective model.
- **Phase 9 - Interpretation:** Translate model and EDA results into food-safety insights, limitations and actionable recommendations.
- **Phase 10 - Communication:** Final visualizations, README, GitHub repository, presentation slides and demonstration.

4. Recommended Repository Structure

```
FINAL_PROJECT/
|
|-- Data/
|   |-- 20250108_opendata_moni_23.csv
|   |-- processed/           # cleaned/model-ready outputs; large files may stay local
|
|-- notebooks/
|   |-- 01_data_inspection.ipynb
|   |-- 02_data_cleaning.ipynb
|   |-- 03_eda_statistics.ipynb
|   |-- 04_feature_engineering.ipynb
|   |-- 05_modeling_evaluation.ipynb
|
|-- src/
|   |-- data_loading.py
|   |-- data_cleaning.py
|   |-- features.py
|   |-- visualization.py
|   |-- modeling.py
|
|-- outputs/
|   |-- figures/
|   |-- tables/
|   |-- models/
|
|-- presentation/
|   |-- project_slides.*
|
|-- .gitignore
|-- environment.yml
|-- requirements.txt
|-- README.md
|-- config.py                # paths/configuration if useful
```

5. Initial Research-Question Framework

The exact questions must be finalized only after inspecting the BVL fields and determining how non-compliance is represented. The project should nevertheless aim for multiple focused questions, for example: Which food/product categories show higher rates of adverse findings? Which sample or monitoring characteristics are associated with non-compliance? Can available pre-outcome characteristics predict elevated failure risk? These are provisional and must be validated against the actual data dictionary and variables.

6. Modeling Standard

Do not choose models merely to satisfy a number. Start with a simple interpretable baseline and compare at least three appropriate classifiers. Candidate models may include Logistic Regression, Decision Tree, Random Forest/Gradient Boosting, and KNN if appropriate for the final feature space. Use train/test separation, cross-validation, justified preprocessing and hyperparameter tuning. Select evaluation metrics based on the food-safety business cost of false negatives and false positives.

7. Git & Documentation Routine

Commit throughout development, not only at submission. Each project day should contain at least one precise atomic commit. Keep the raw 1.17 GB CSV out of GitHub unless an appropriate large-file strategy is explicitly chosen. The README should explain the business problem, dataset/source, methods, repository structure, setup instructions, results, limitations and recommendations.

8. Immediate Next Steps

1. Initialize the Git repository and create .gitignore.

2. Create the professional folder structure above.
3. Make the first atomic commit for environment/project setup.
4. Create 01_data_inspection.ipynb.
5. Safely inspect a small sample of the 1.17 GB BVL CSV before deciding how to load/process the full dataset.
6. Use the observed fields to finalize the business problem, research questions and target variable.

Guiding principle: Every major analytical choice should be explainable: what was done, why it was appropriate, what problem it solves, and how it affects the analysis or model.